Ques. WHAT ARE THE DIFFERENCES BETWEEN FLASH MEMORY AND EEPROM?

S. Dwivedi

Ans. Flash memory and electrically-erasable programmable read-only memory (EEPROM) are both electrically-writable and erasable programmable ROMs (PROMs), and are used in microcontrollers (MCUs) as on-chip memories. However, differences between these are with respect to configuration, applications, number of erase/write cycles, endurance and data retention periods.

Configuration. Flash memory has a simpler structure and more data storage capacity. It is a block-wise erasable memory, whereas EEPROM is a byte-wise erasable memory. Flash memory is designed for high speed and high density at the expense of large erase blocks. A block contains many pages as shown in Fig. 1.

EEPROM is a non-volatile memory with small erase blocks. Flash memory can, therefore, provide a larger memory capacity. In MCUs, on-chip flash memory is several tens to thousands of bytes, whereas on-chip EEPROM is just several tens of bytes.

Write function. In flash memory, data can be written from an erased state (for example, changing from 1 to 0), but not in the opposite direction (from 0 to 1). Data must be erased in block units.

In EEPROM, old data can be over-written simply by writing new data to the required address.

Number of rewrite cycles. Flash memory has fewer rewrite cycles, up to 10,000 depending on the actual product. EEPROM usually guarantees at least one million rewrites.

Storage. Flash memory is used as ROM for storing programs or large-capacity (static or semi-static) data in applications where data does not need to be overwritten frequently. EEPROM is used as an extension of RAM to store a small amount of data when power is turned off in applications where data is overwritten more frequently.

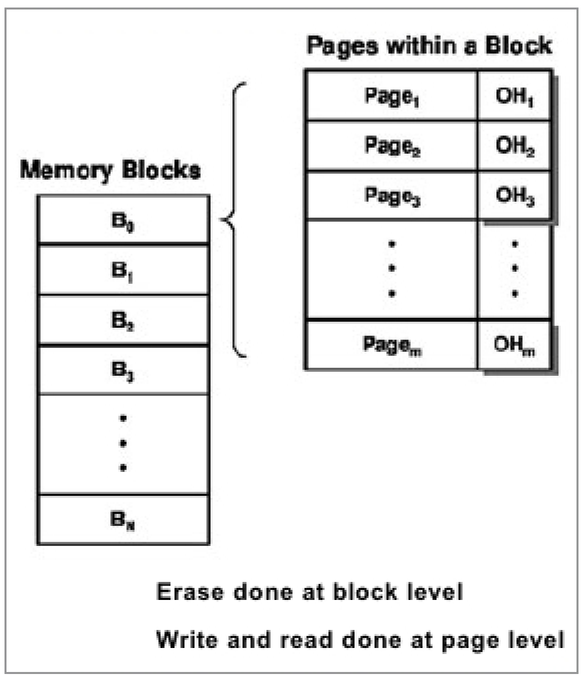

Fig. 1: Simple flash memory

Q2. WHAT IS AN OPTICAL FIBRE IN ELECTRONICS? HOW IS DATA CONVERTED INTO A SERIES OF LIGHT PULSES IN OPTICAL-FIBRE COMMUNICATION?

Balakumaran

A2. Optical-fibre communication is a method of transmitting information (analogue or digital) from one place to another, by sending pulses of light through an optical fibre. Block diagram of a typical optical-fibre system is shown in Fig. 2.

It is similar to the copper wire system. The difference is that fibre optics use light pulses to transmit information down the cable fibres instead of using electronic pulses to transmit information down the copper lines.

It is mostly used in telephones, Internet communication and cable televisions. Sound, video and data signals are electronically digitised into a stream of on and off pulses, and converted into an equivalent electric signal.

Suppose, you want to transmit a sound signal over a fibre-optic cable. First, convert the sound/analogue signal to digital signal using a suitable converter. A light source such as LED or laser can be used for generating light pulses, which move easily down the fibre-optic line because of total internal reflection.

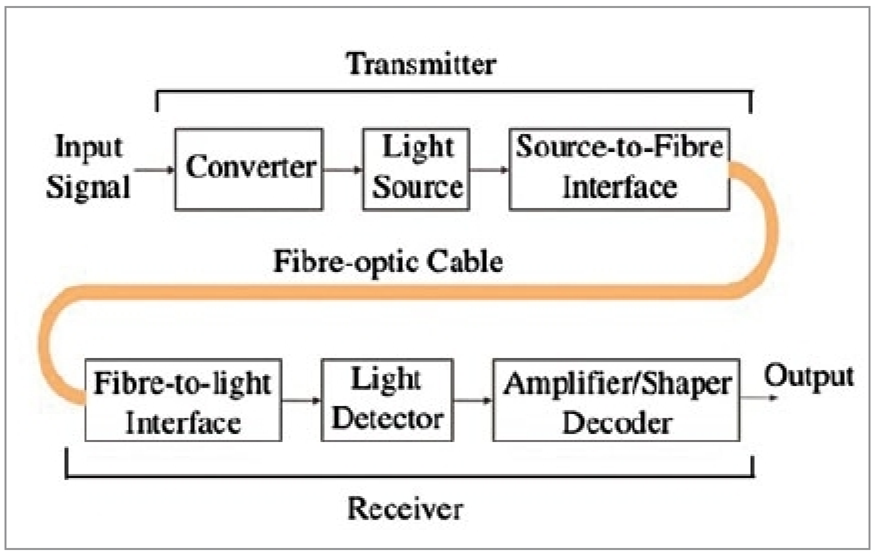

Fig. 2: Block diagram of fibre-optical system

Lasers generate modulated light signals and transmit these through the optical system by turning each laser device on and off. Device on represents digital 1, and off digital 0.

Modern fibre systems with a single laser can turn on and off several billions of times per second. Multiple lasers with different colours can fit multiple signals into the same fibre cable.

Each optical signal has its own wavelength or frequency. A typical optical network consists of six different sections including optical transmitter, multiplexer, optical amplifier, switching/routing, de-multiplexer and receiver.

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